

Introduction

Jumping spiders possess a unique visual system that makes them well-suited to cognition studies. While their large principal eyes have impressive visual resolution for their body size, they have an extremely narrow field of view, and cannot perceive movement. Thus, they rely on their smaller secondary eyes to inform them where to focus their gaze. This makes them simultaneously very selective in what they pay attention to, while still being aware of all possibilities in front of them at once [1].

Using a retina tracker (Figure 1) to follow the movements of their principal eyes, we can record what kinds of visual stimuli are more likely to attract their attention on a moment-by-moment basis, giving us an opportunity for insight into the cognitive processes that control this behavior [2].



Figure 1a: Spider secured in painted Parafilm for eye tracking

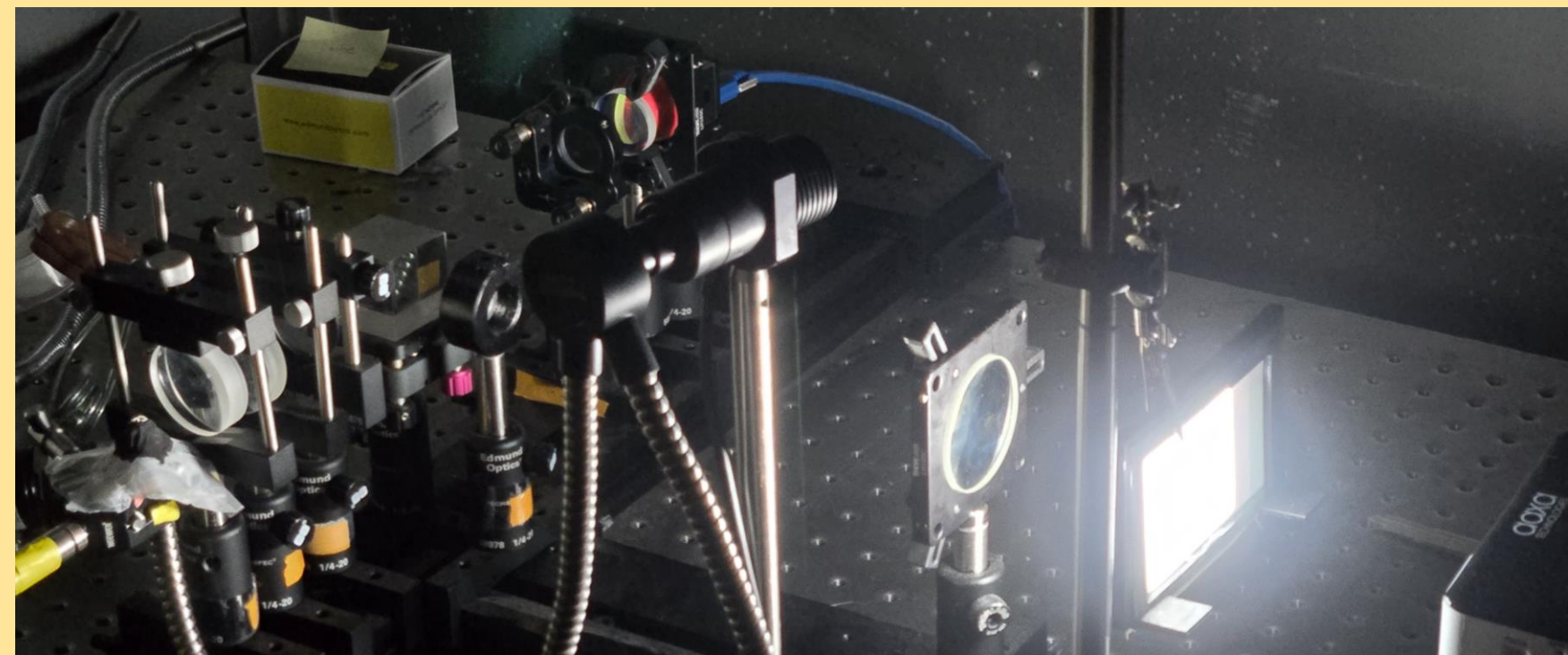


Figure 1b: Eye tracker setup (clockwise from left: spider on top of IR fiber optic cable; IR camera recording retina position; projector and screen displaying stimulus)

The goal of this experiment was to investigate how eye tracking could be used to interpret jumping spiders' documented capacity for associative learning [3]. To this end, I wanted to find out whether they might begin to anticipate the location of an unrevealed positive stimulus by spending more time looking at a neutral one that is associated with it. If successful, this general method could be applied to research into a variety of different forms of visual learning, from pattern recognition to numerosity.

METHODS & MATERIALS

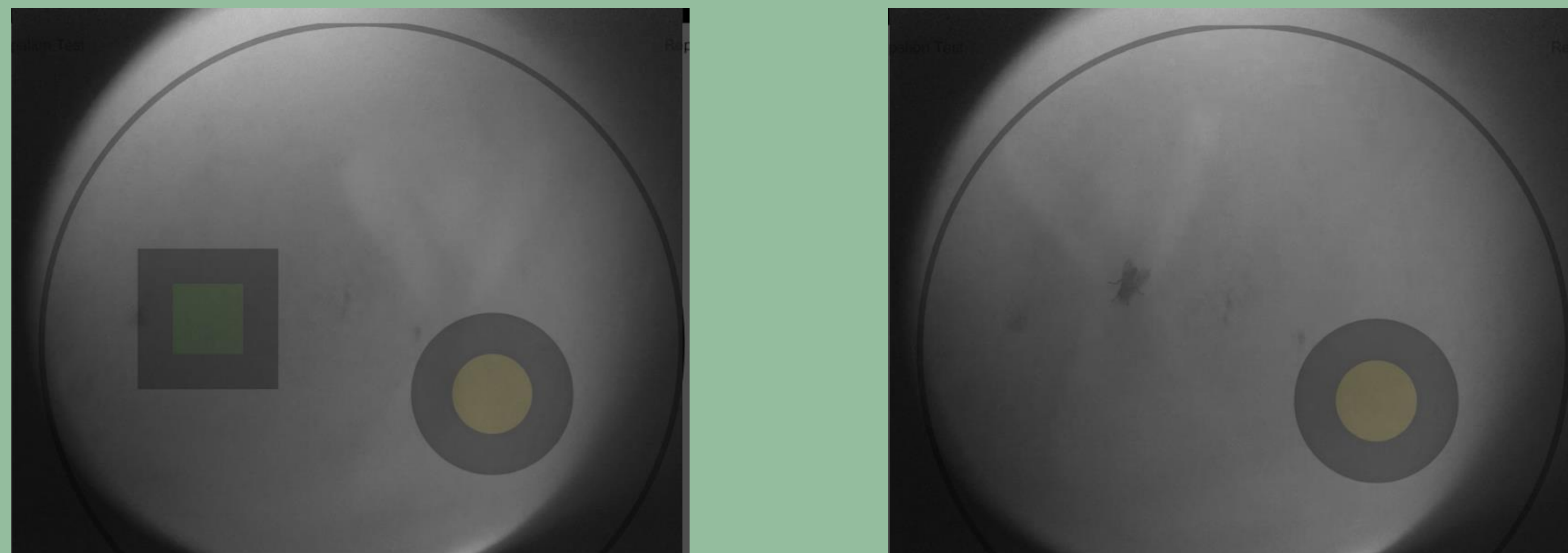


Figure 2: Stimulus design; The spiders' retinæ appear as a light gray "X" when focused on a target

I used 10 locally-caught bold jumping spiders, *phidippus audax*, for my sample group, secured to the eye tracker surrounded by a black plastic sheet to reduce distractions. Recording their retina movements, I presented them each with 20 consecutive repetitions of the following stimulus, as seen in Figure 2: Two different objects appeared in a randomized position for 10 seconds, after which one of them was replaced by a video of a moving fly for 6.5 seconds, while the other stayed in place. Then, both were removed for a 10-second rest period before the next trial began.

To control for bias, the object being replaced by the fly was always the square for half of the spiders, and always the circle for the others. Results were scored based on the amount of time the focal point between the retinas was focused on each shape during the 10-second window that both were present, and a linear regression model was used to calculate the difference between looking times at each object (time spent looking at the stimulus with the fly minus time spent looking at the other within each trial, in seconds) by trial number with the slope for each subject as a random effect.

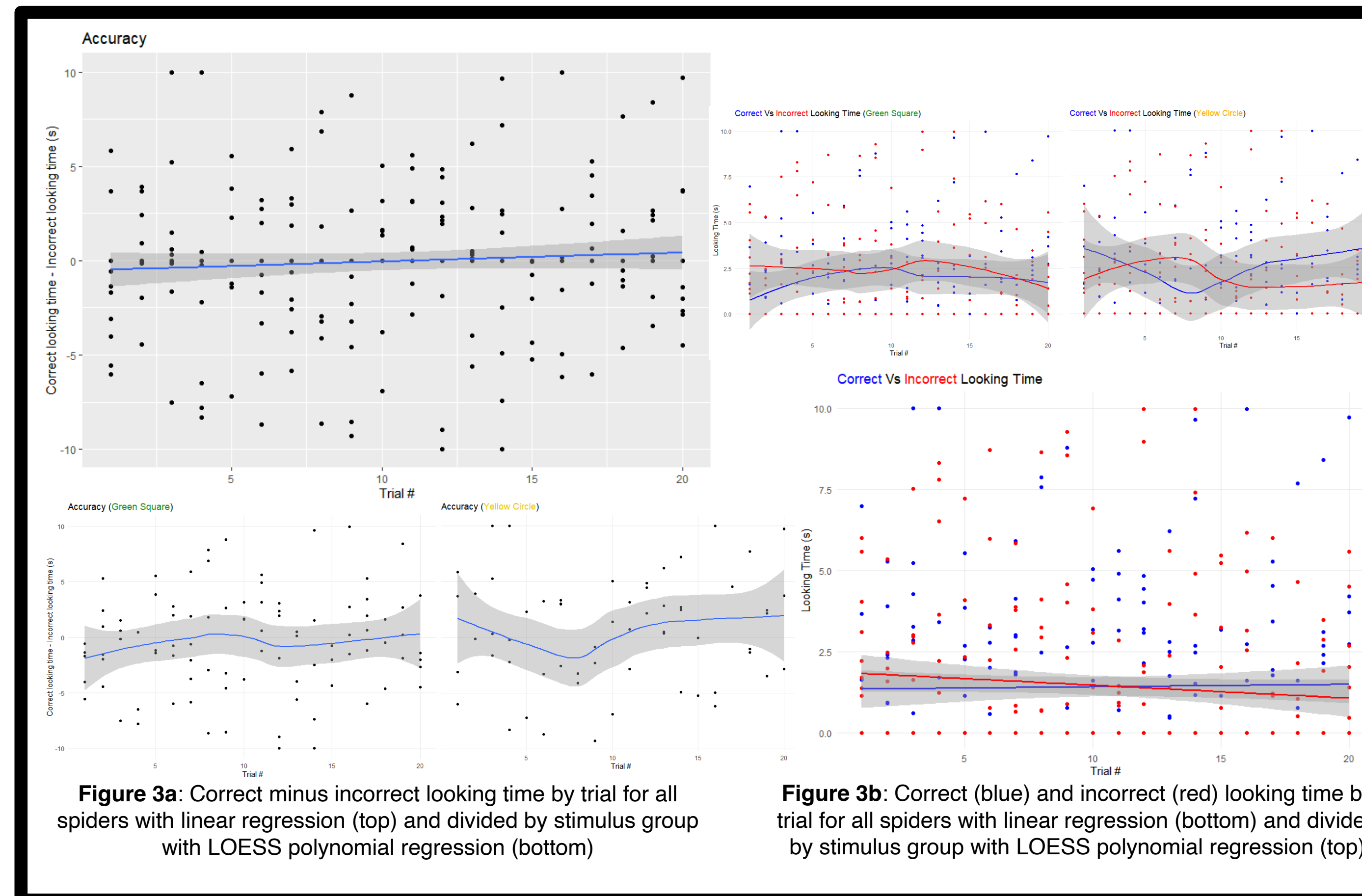


Figure 3a: Correct minus incorrect looking time by trial for all spiders with linear regression (top) and divided by stimulus group with LOESS polynomial regression (bottom)

Figure 3b: Correct (blue) and incorrect (red) looking time by trial for all spiders with linear regression (bottom) and divided by stimulus group with LOESS polynomial regression (top)

RESULTS

- Overall, although the difference between correct and incorrect looking times across all spiders did nominally increase between the first and last trial, the results were far from significant, increasing by just 0.048 seconds per trial with an R^2 value of 0.014 (Figure 3).
- Though change over time was not significant, bias between shapes was. Among trials in which they were looked at, mean looking times for the circle and the square were 5.5 and 4.1 seconds respectively, with a p-value of $2.2e-16$ and t-value of 13.65 when compared with an unpaired t-test (Figure 4).
- One spider performed noticeably better than others, with difference in correct vs incorrect looking times increasing by 0.29 seconds per trial and a p-value of 0.178, though R^2 remained low at 0.1 (Figure 5).

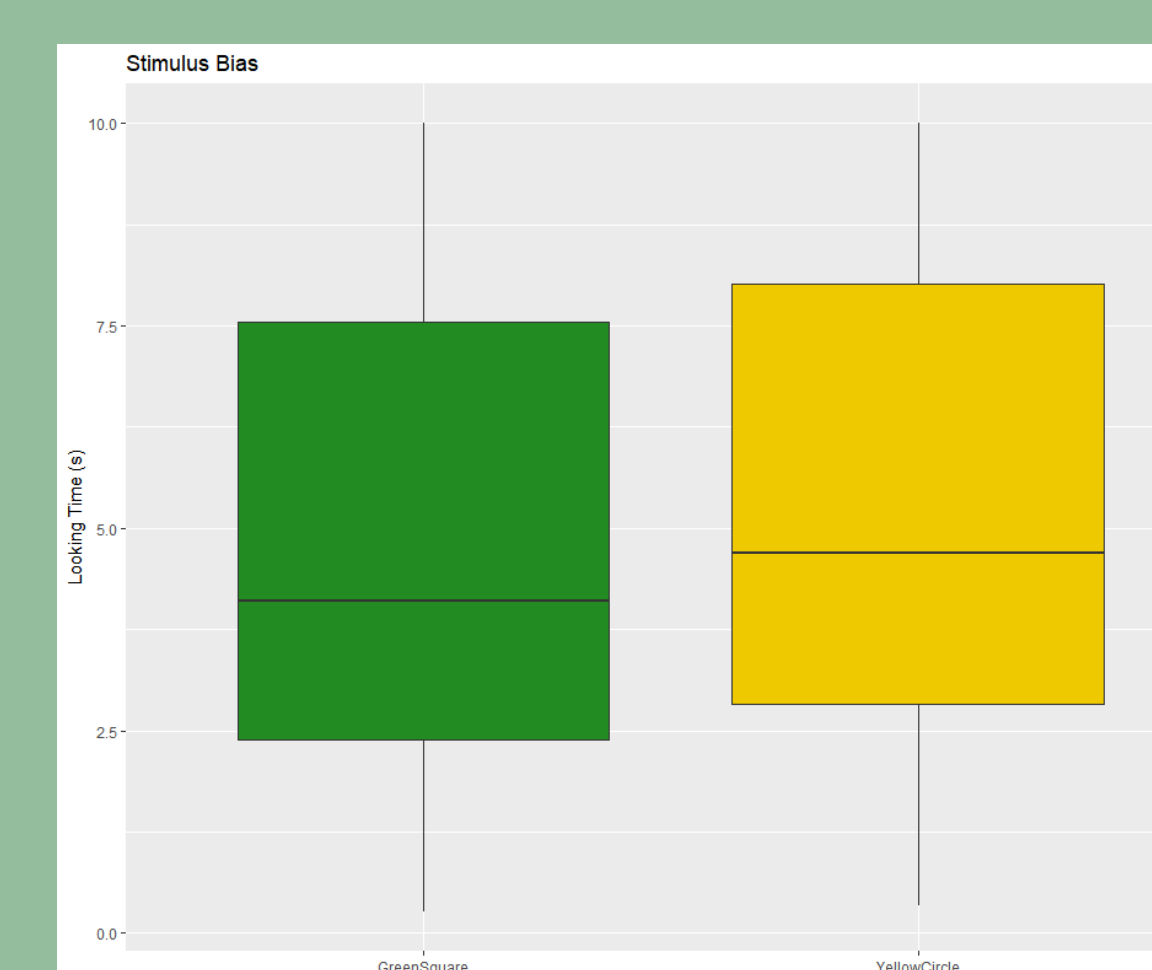
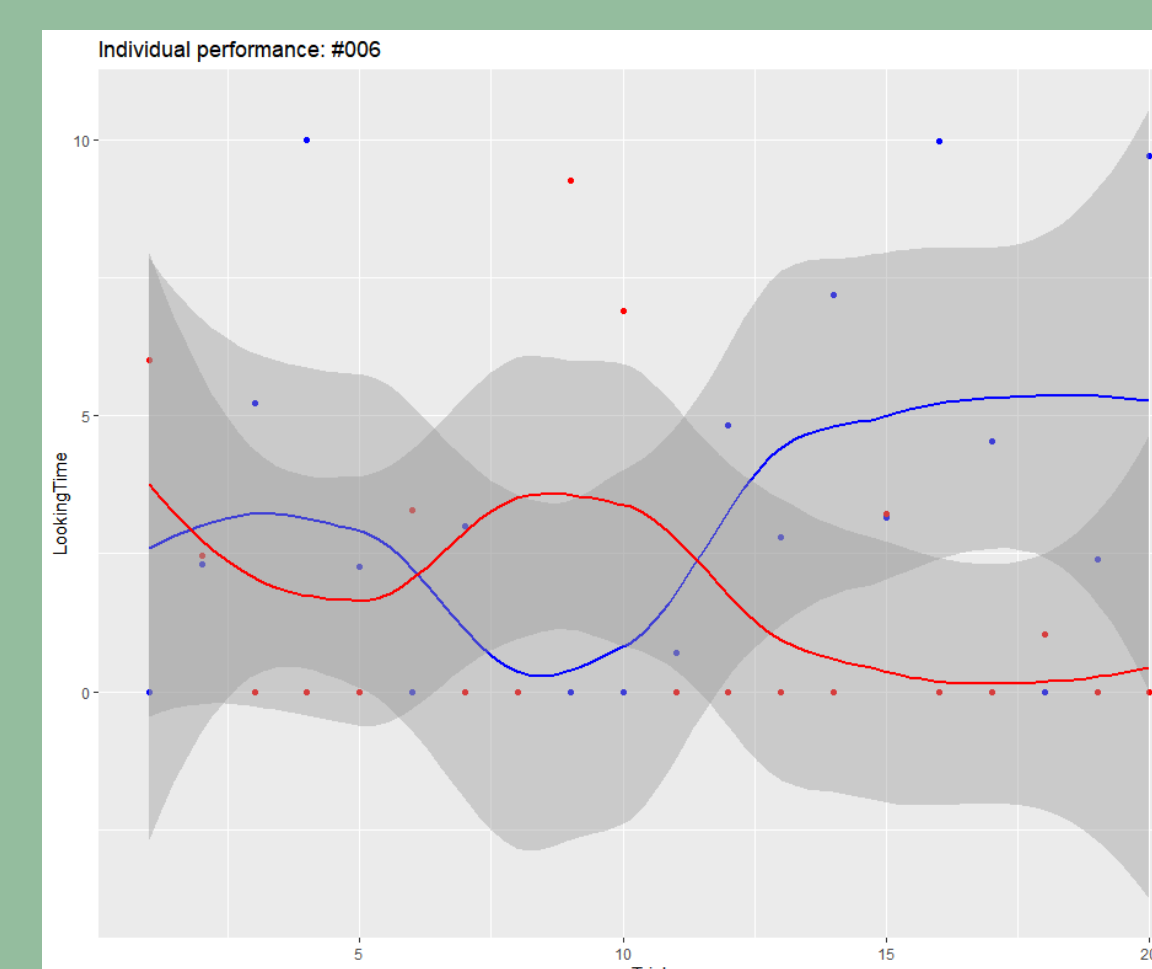
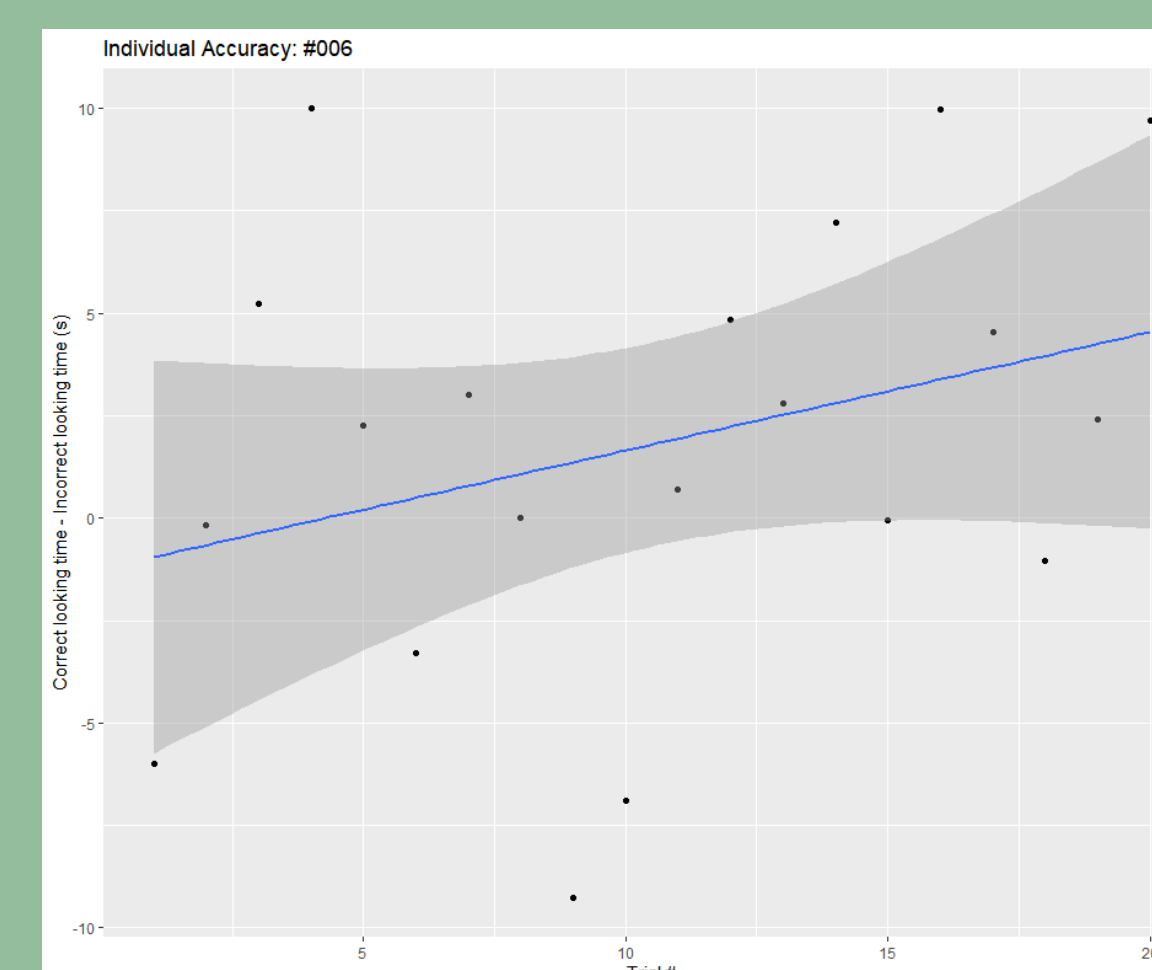


Figure 4: Spiders overall looked longer at the yellow circle than they did the green square.

Figure 5: Spider #006 not only had the highest total looking time (100.3 seconds) and the greatest improvement across trials, it was also the closest to having a statistically significant individual result (above left). A LOESS regression told a promising story of alternating interest, ending with a preference for the correct stimulus that briefly rose beyond the 95% confidence interval (left).

However, being in the group trained on the circle could have given this individual an advantage, and the amount of variation in looking time is still too high to conclusively prove that its performance was not due to chance.

DISCUSSION

- My results strongly supported the null hypothesis that jumping spiders did not pay more attention to a particular stimulus when conditioned in this way. In general, looking times at both shapes were largely random and highly variable across trials, with no sign of significant improvement.
- Stimulus preference could have played a significant role in the data that was collected. Although the same number of spiders were trained on each shape, they appeared to react differently to the green square than they did to the yellow circle. It's conceivable that the color green is interpreted by them as distant leaves or open background [4], prompting them to unfocus their retinæ in an attempt to look into a more distant focal plane and gauge depth.
- There are many possible interpretations of what constitutes "looking," and different scoring systems would likely yield different results. The aimless gaze of an uninterested spider might drift through the center of a stimulus by pure chance, while some spiders appeared to strategically use the tips of their retinæ to observe both at once without focusing on either (Figure 6).

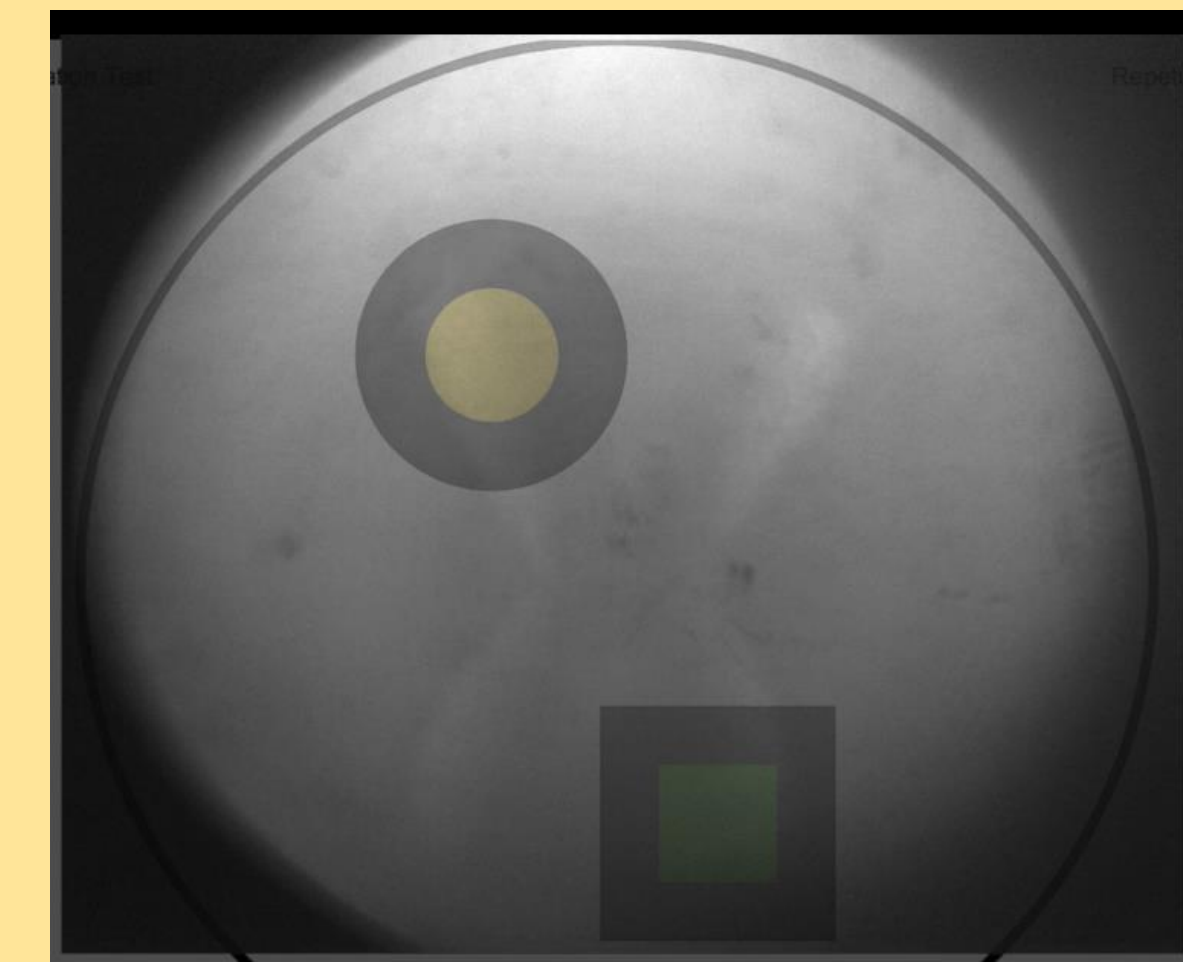
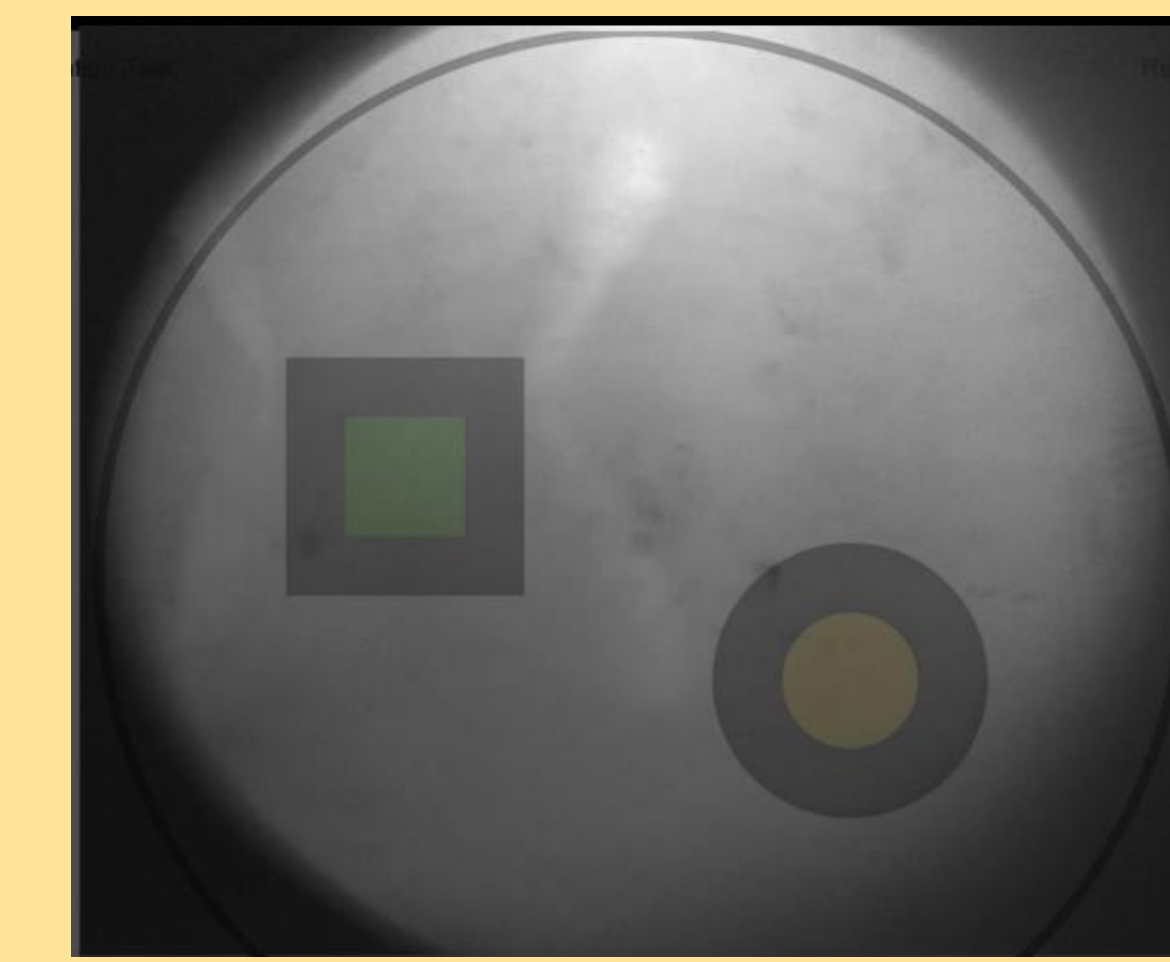


Figure 6: Some spiders seemed to unfocus their eyes in response to looking at a stimulus (left) or preferred to look at a spot directly between them (right)

- Much of the noise in the data came from variance between which stimulus was looked at first by the spider, especially when one of the shapes appeared where its retinæ were already pointed. Longer trial periods would have given the spiders more time to change their focus, allowing for more nuance at the upper end of looking times. Reducing the size of the stimuli would both lower the chance of it appearing in the spider's initial field of view, and better distinguish when they were purposefully looking at it or not.
- Spiders are expected to differ not only in performance, but in learning time, initial preferences, and points at which they simply lose interest [5], among other factors, and many of these subtleties are lost in an aggregate linear regression. A polynomial regression such as LOESS may be better at visualizing these types of patterns, but much more data would be needed to describe them with any degree of confidence across individuals.
- Performance varied considerably between subjects, but one out of the twelve came closer to succeeding than the others. If the design is marginally approachable to only the top percentile of spiders, it may be that the design is within their capability, but simply too difficult for the majority in its current form.

References

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